KISHINCHAND CHELLARAM COLLEGE, MUMBAI - 20
T.Y. B.Sc. I.T. Semester VI

Practical 9: Configure OSPF MD5 authentication and support SSH connections.

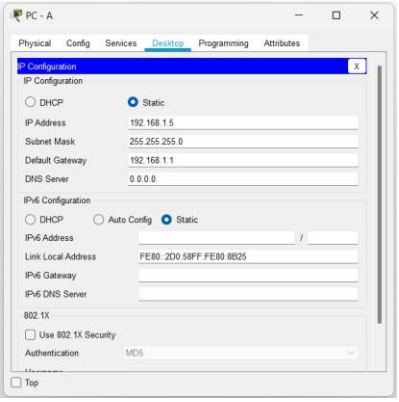
A) Topology

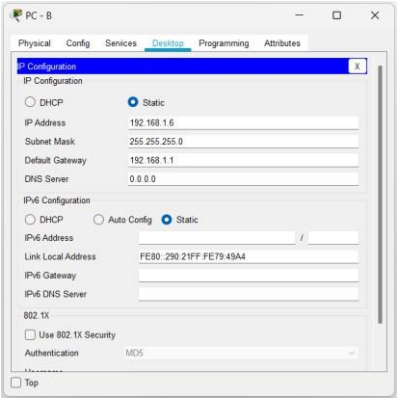
1. Drag and drop the required icons that are 3 1941 Routers; 1 PT-PC; 2 2960-24TT Switches; 2 PT-Servers and rename them accordingly.

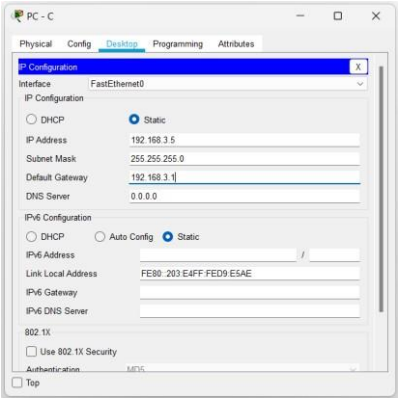
2. Define the respective IP configuration by going to Desktop and then to IP Configuration for PC – A (Server), PC - B(Server) and PC - C.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1	S1 F0/6
PC-B	NIC	192.168.1.6	255.255.255.0	192.168.1.1	S2 F0/18
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1	S3 F0/18

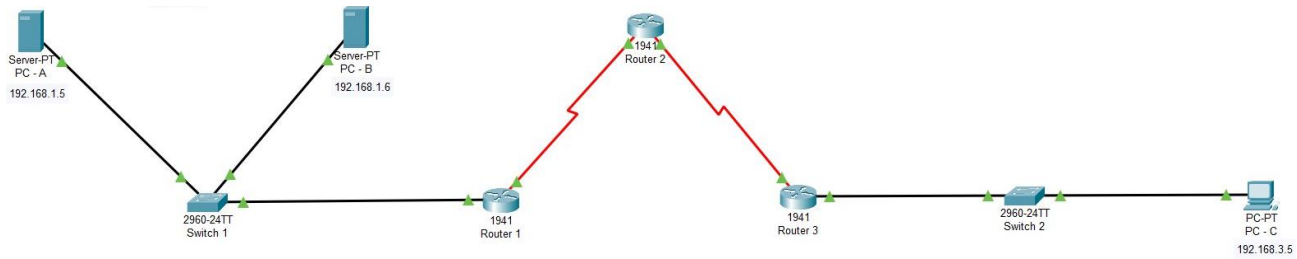






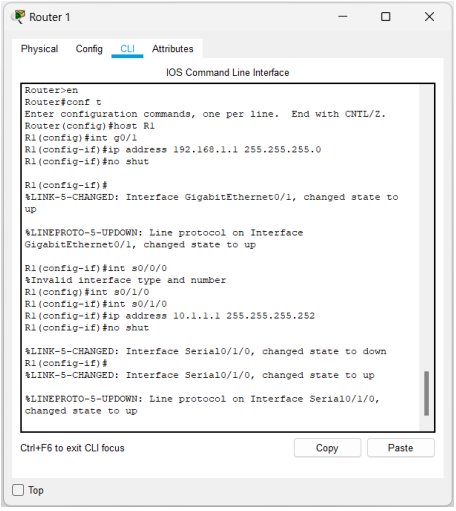
3. Add serial ports to all the three routers by going to Physical of Router 1, switching the router off and adding HWIC-2T to both the empty slots and turn the switch on, do the same for Router 2 and Router 3.

4. Connect the devices with the respective wired connection.

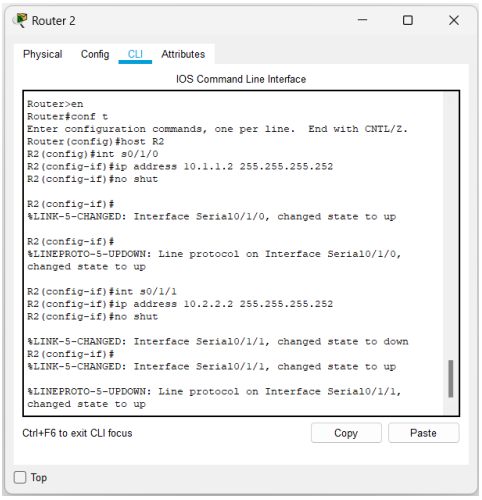


B) Configure Router

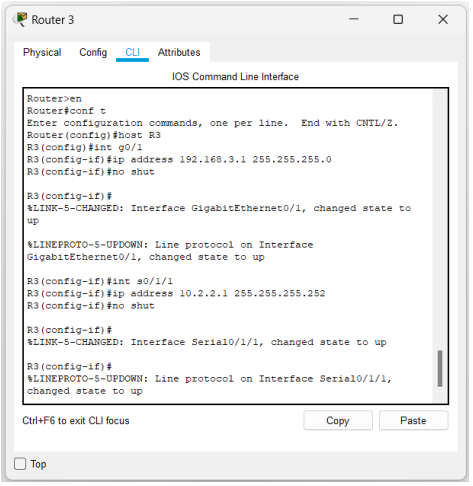
<u>Router 1</u>
Router>en
Router#conf t
Router(config)#host r1
r1(config)#int g0/1
r1(config-if)# ip address 192.168.1.1 255.255.255.0
r1(config-if)#no shut
r1(config-if)#int s0/0/0
r1(config-if)#ip address 10.1.1.1 255.255.255.252
r1(config-if)#no shut



<u>Router 2</u>
Router>en
Router#conf t
Router(config)#host r2
r2(config)#int s0/0/0
r2(config-if)#ip address 10.1.1.2 255.255.255.252
r2(config-if)#no shut
r2(config-if)#int s0/0/1
r2(config-if)#ip address 10.2.2.2 255.255.255.252
r2(config-if)#no shut



<u>Router 3</u>
Router>en
Router#conf t
Router(config)#host r3
r3(config)#int g0/1
r3(config-if)#ip address 192.168.3.1 255.255.255.0
r3(config-if)#no shut
r3(config-if)#int s0/0/1
r3(config-if)#ip address 10.2.2.1 255.255.255.252
r3(config-if)#no shut



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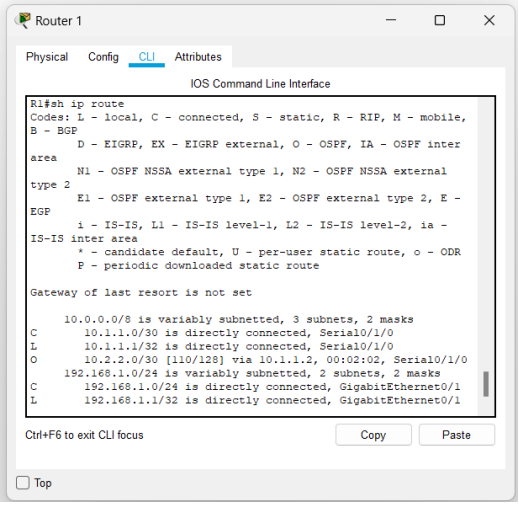
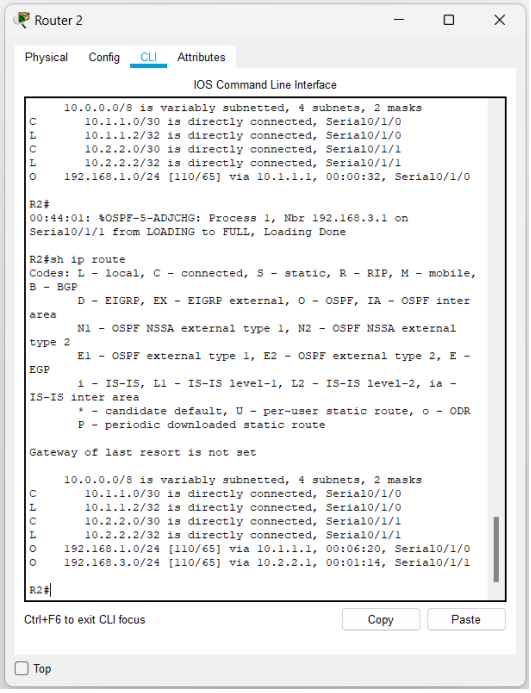
C) Configure OSPF

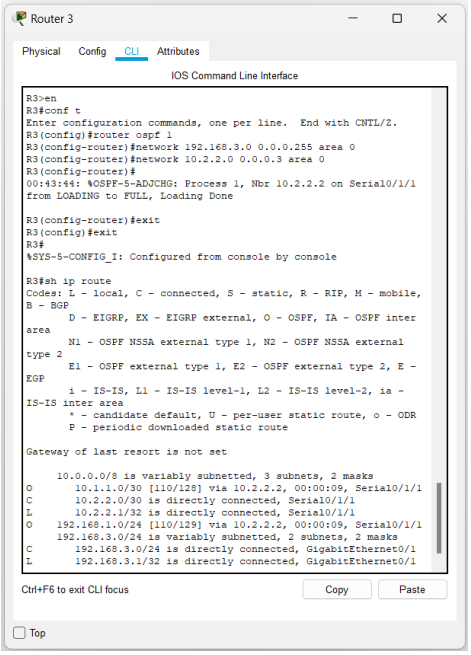
Router 1
r1(config)#router ospf 1
r1(config-router)#network 192.168.1.0 0.0.0.255 area 0
r1(config-router)#network 10.1.1.0 0.0.0.3 area 0
r1(config-router)#exit
r1(config)#exit
r1#sh ip route



Router 2
r2(config)#router ospf 1
r2(config-router)#network 10.1.1.0 0.0.0.3 area 0
r2(config-router)#network 10.2.2.0 0.0.0.3 area 0
r2(config-router)#exit
r2(config)#exit
r2#sh ip route



Router 3
r3(config)#router ospf 1
r3(config-router)#network 192.168.3.0 0.0.0.255 area 0
r3(config-router)#network 10.2.2.0 0.0.0.3 area 0
r3(config-router)#exit
r3(config)#exit
r3#sh ip route

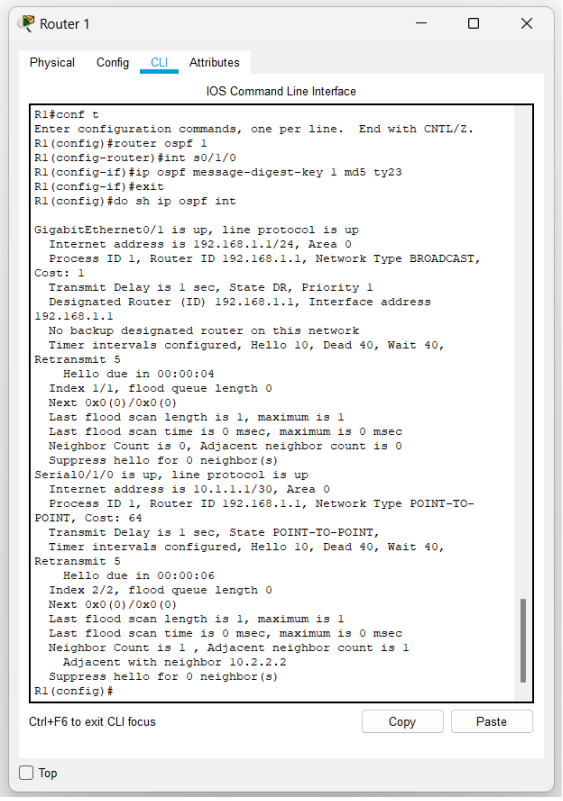


The O proves that OSPF was successful.

D) Configure MD5 Authentication

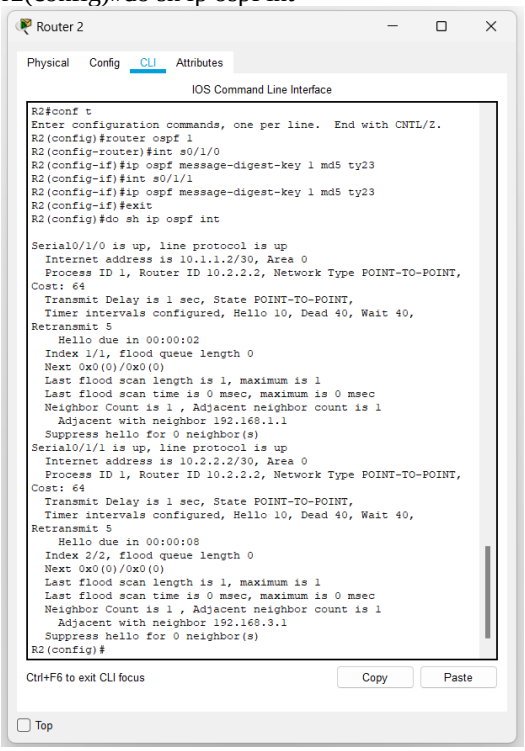
Router 1

r1(config)#router ospf 1
r1(config-router)#int s0/0/0
r1(config-router)#ip ospf message-digest-key 1 md5 ty23
r1(config-router)#exit
r1(config)#do sh ip ospf int



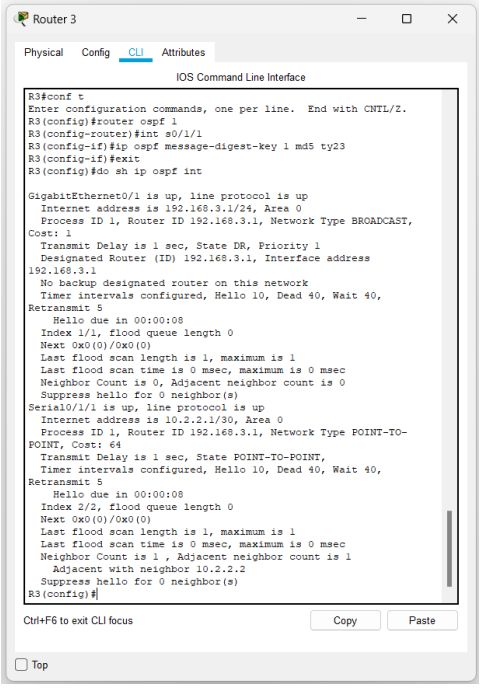
Router 2

r2(config)#router ospf 1
r2(config-router)#int s0/0/0
r2(config-router)#ip ospf message-digest-key 1 md5 ty23
r2(config-router)#int s0/0/1
r2(config-router)#ip ospf message-digest-key 1 md5 ty23
r2(config-router)#exit
r2(config)#do sh ip ospf int



Router 3

r3(config)#router ospf 1
r3(config-router)#int s0/0/1
r3(config-router)#ip ospf message-digest-key 1 md5 ty23
r3(config-router)#exit
r3(config)#do sh ip ospf int

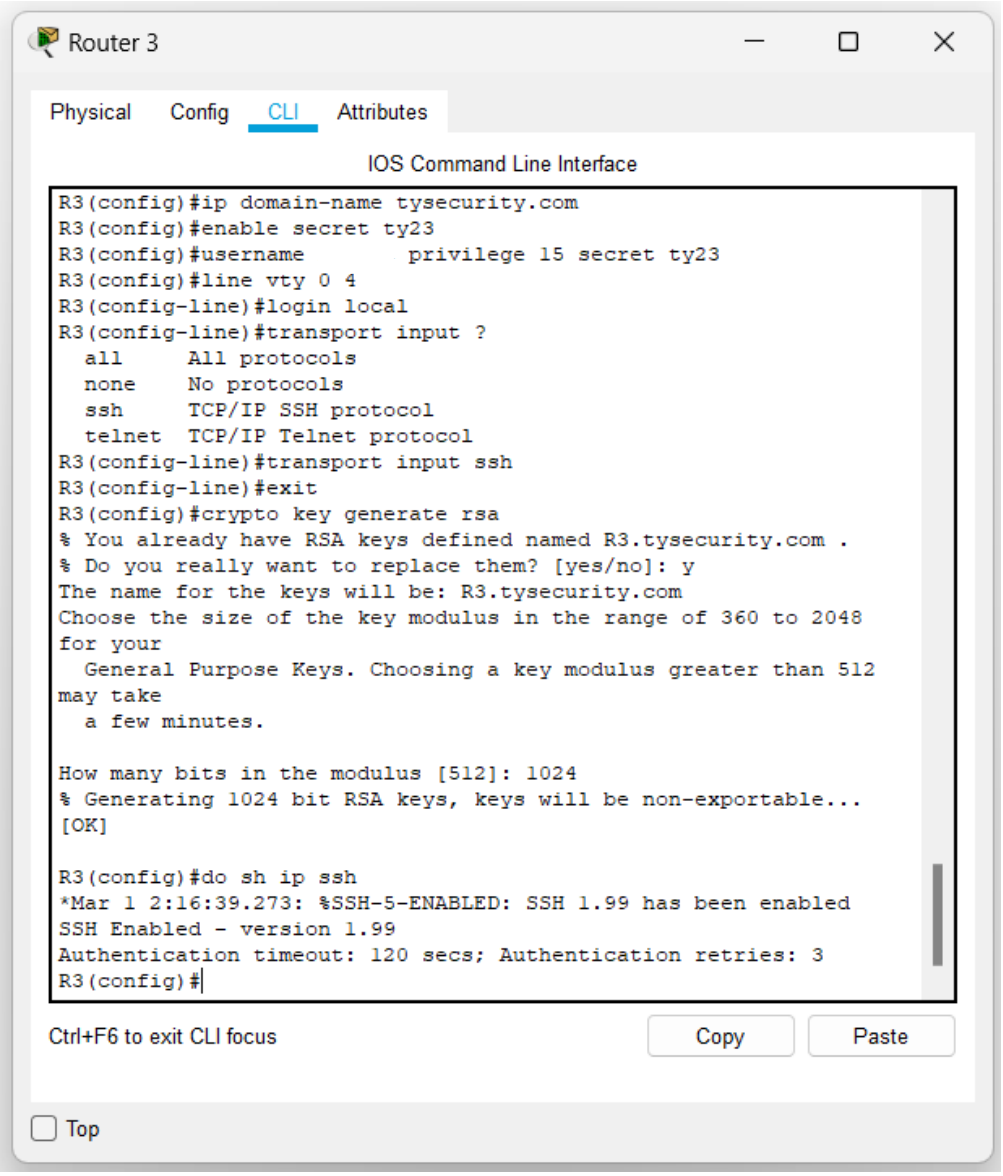


E) Configure Router 3 to support SSH connections

Router 3
r3(config)#ip domain-name tyitsecurity.com
r3(config)#enable secret ty23

Exit the connections and enable the router again. It will ask for password, enter ty23, it will not be visible but the system will accept it. Once you press enter, it will take you to config mode of the router.

r3(config)#username admin123 privilege 15 secret
ty23r3(config)#line vty 0 4
r3(config-line)#login local
r3(config-line)#transport input ssh
r3(config-line)#exit
r3(config)#crypto key generate rsa
How many bits in the modulus? 1024
r3(config)#do sh ip ssh



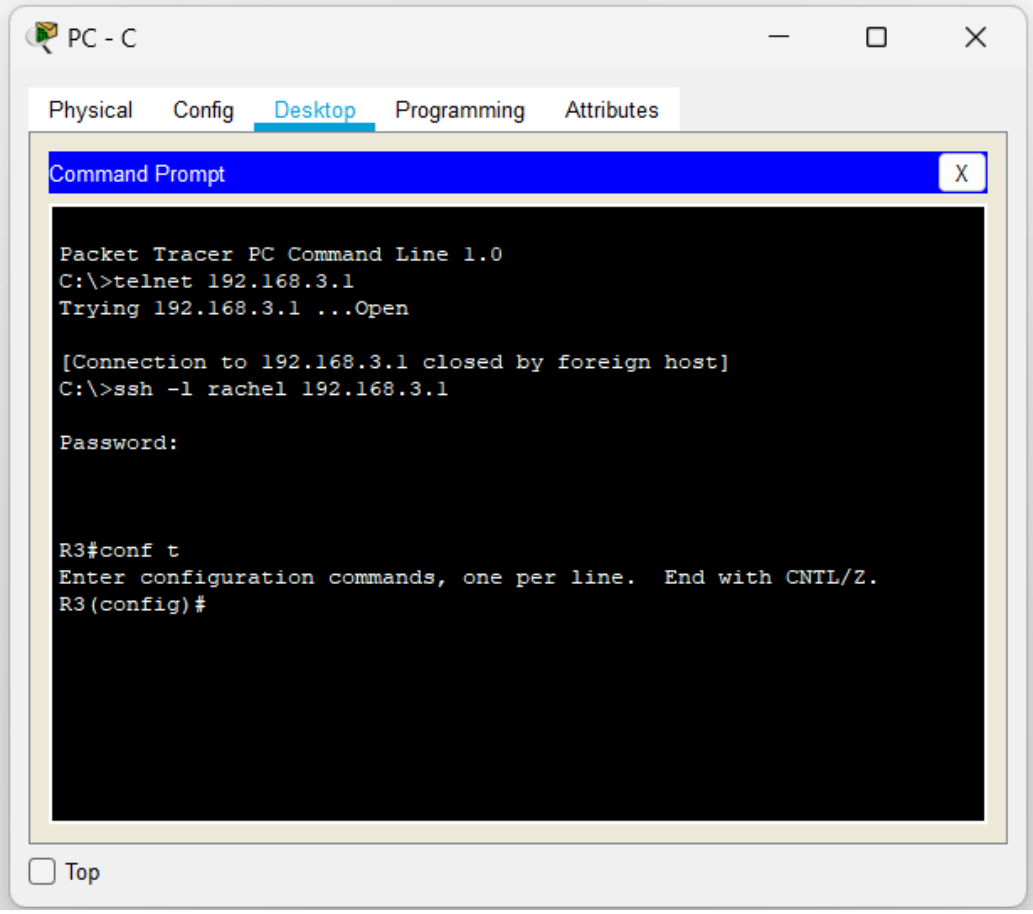
F) Check results

Open Command Prompt of PC – C.

Enter telnet 192.168.3.1, the connection will open.

Enter ssh -l admin123 192.168.3.1

Password will be asked type ty23 which will not be visible but will be accepted by the system. Router 3 configuration is now accessible from PC - C.



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Practical 10: Configure AAA authentication on routers.

A) Topology

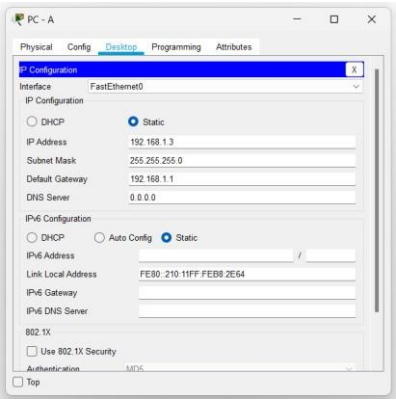
Drag and drop the required icons; rename them and change the IP configuration and then connect them.

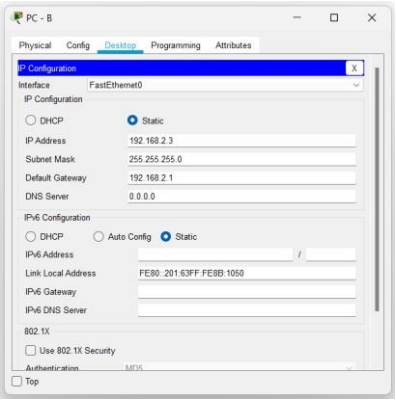
1. Drag and drop the required icons that are 3 1941 Routers; 3 PT-PCs'; 3 2960-24TT Switches; 2 PT-Servers and rename them accordingly.

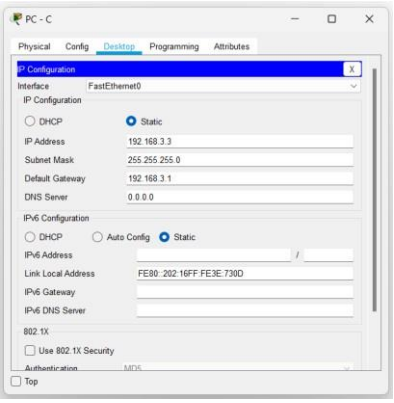
2. Define the respective IP configuration by going to Desktop and then to IP Configuration for PC - A, PC – B, PC – C, TACACS+ Server and RADIUS Server.

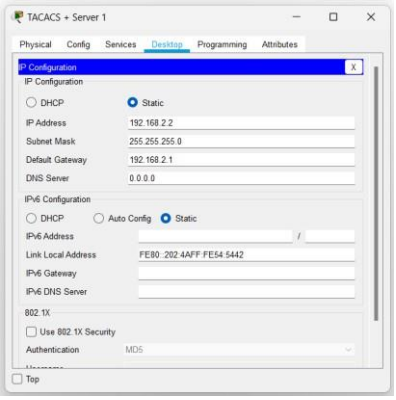
Addressing Table

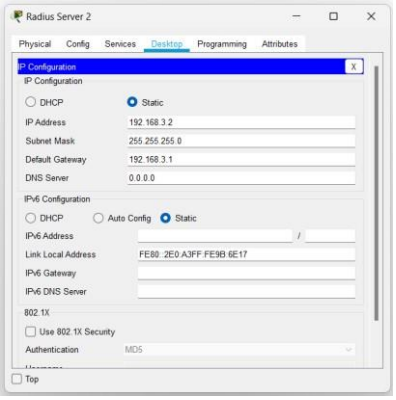
Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A	S2 F0/2
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.1	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
TACACS+ Server	NIC	192.168.2.2	255.255.255.0	192.168.2.1	S2 F0/6
RADIUS Server	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/2
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 F0/1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18





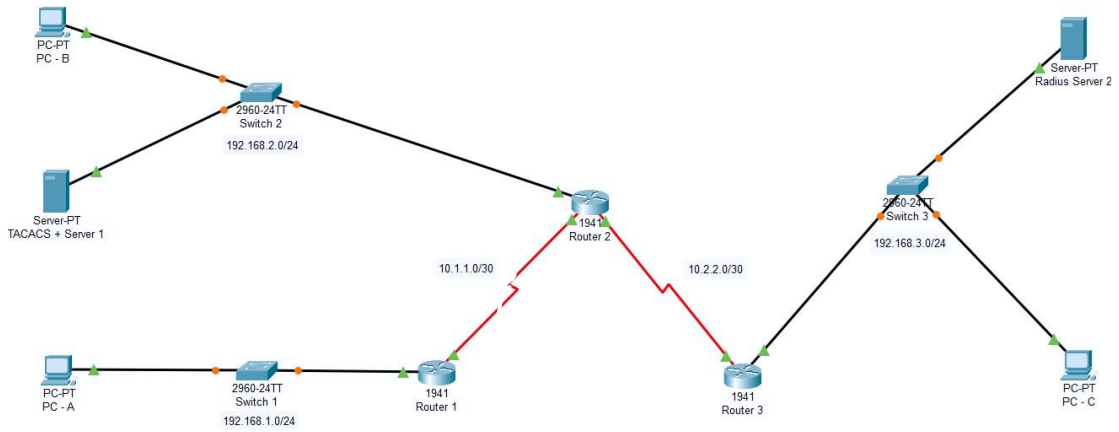






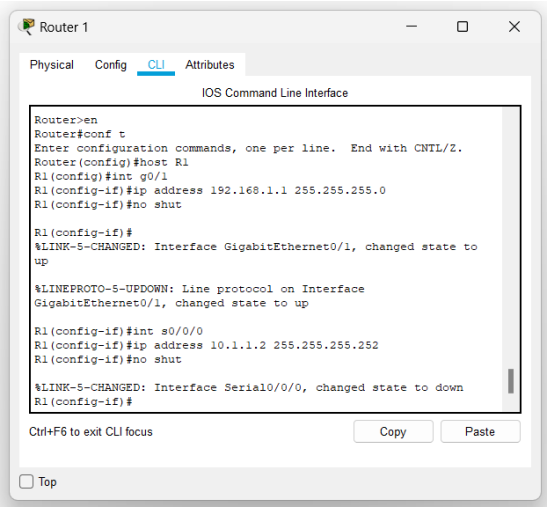
3. Add serial ports to all the three routers by going to Physical of Router 1, switching the router off and adding HWIC-2T to both the empty slots and turn the switch on, do the same for Router 2 and Router 3.

4. Connect the devices with the respective wired connection.

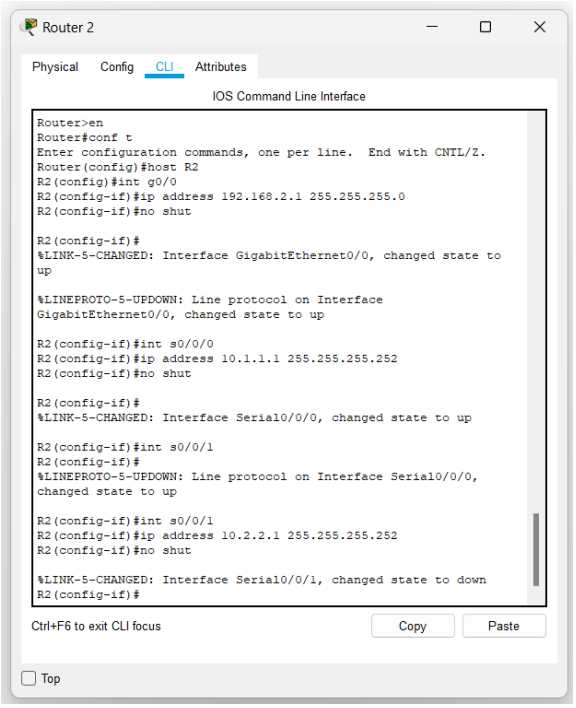


B) Configure Router

<u>Router 1</u>
Router>en
Router#conf t
Router(config)#host r1
r1(config)#int g0/1
r1(config-if)# ip address 192.168.1.1 255.255.255.0
r1(config-if)#no shut
r1(config-if)#int s0/0/0
r1(config-if)#ip address 10.1.1.2 255.255.255.252
r1(config-if)#no shut

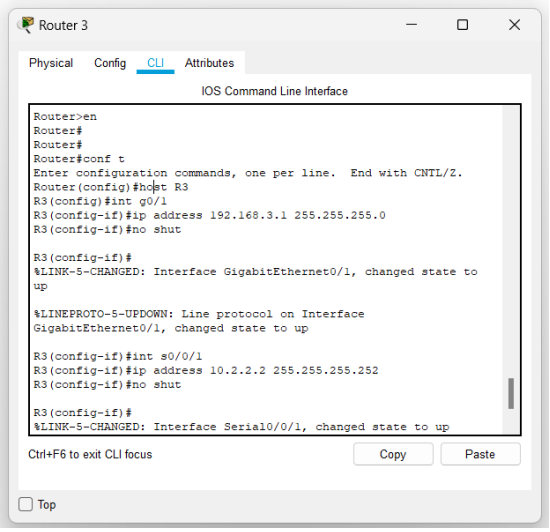


<u>Router 2</u>
Router>en
Router#conf t
Router(config)#host r2
r2(config)#int g0/0
r2(config-if)#ip address 192.168.2.1 255.255.255.0
r2(config-if)#no shut
r2(config-if)#int s0/0/0
r2(config-if)#ip address 10.1.1.1 255.255.255.252
r2(config-if)#no shut
r2(config-if)#int s0/0/1
r2(config-if)#ip address 10.2.2.1 255.255.255.252
r2(config-if)#no shut



Router 3

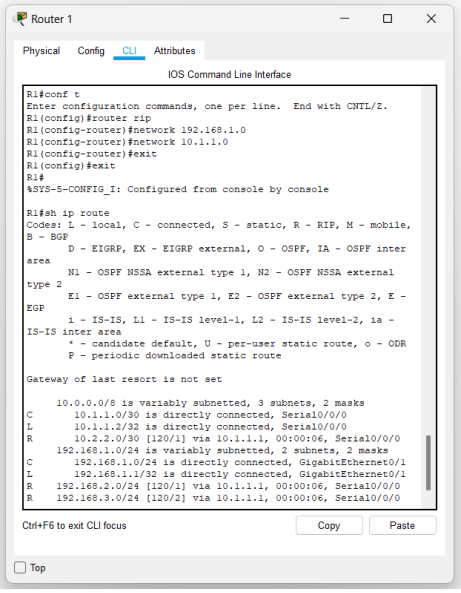
Router>en
Router#conf t
Router(config)#host r3
r3(config)#int g0/1
r3(config-if)#ip address 192.168.3.1 255.255.255.0
r3(config-if)#no shut
r3(config-if)#int s0/0/1
r3(config-if)#ip address 10.2.2.2 255.255.255.252
r3(config-if)#no shut



C) Configure RIP on routers

Router 1

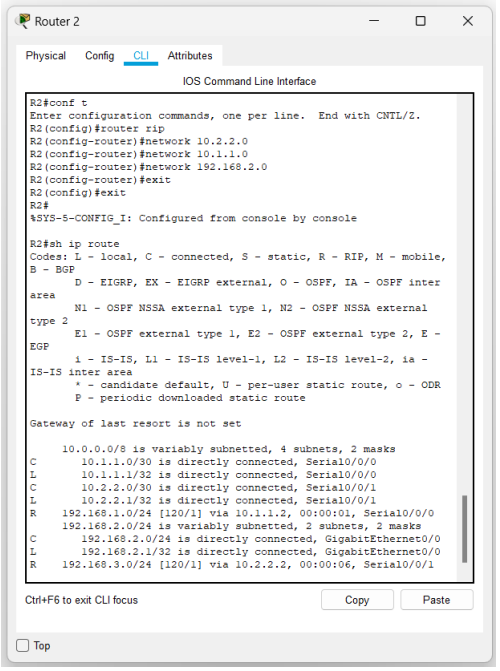
r1(config)#router rip
r1(config-router)#network 192.168.1.0
r1(config-router)#network 10.1.1.0
r1(config-router)#exit
r1>sh ip route



The R proves that RIP was successful.

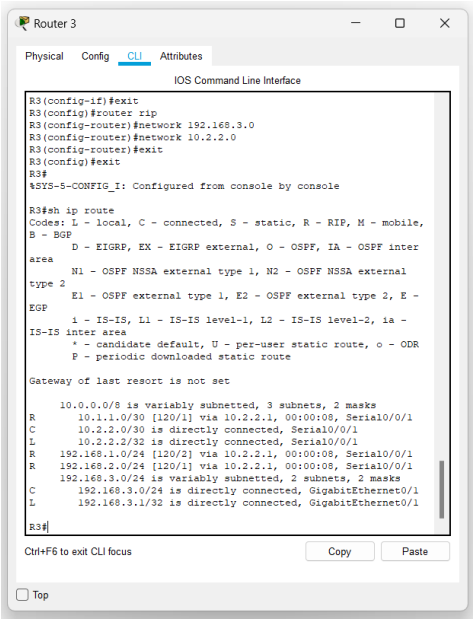
Router 2

r2(config)#router rip
r2(config-router)#network 10.1.1.0
r2(config-router)#network 192.168.2.0
r2(config-router)#network 10.2.2.0
r2(config-router)#exit
r2>sh ip route



Router 3

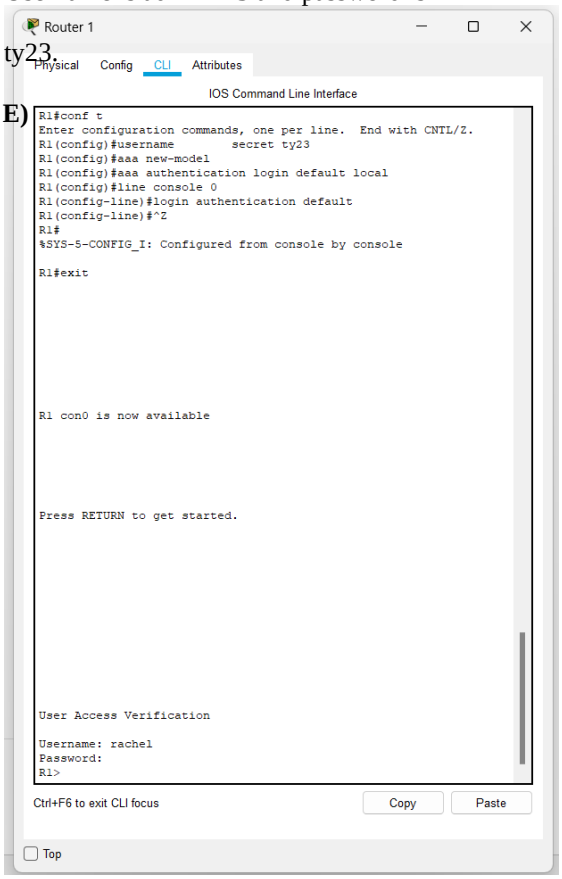
r3(config)#router rip
r3(config-router)#network 192.168.3.0
r3(config-router)#network 10.2.2.0
r3(config-router)#exit
r3>sh ip route



D) Configure local AAA authentication on Router 1

Router 1
r1>en
r1#conf t
r1(config)#username admin123 secret ty23r1(config)#aaa new-model
r1(config)#aaa authentication login default local
r1(config)#line console 0
r1(config-line)#login authentication default

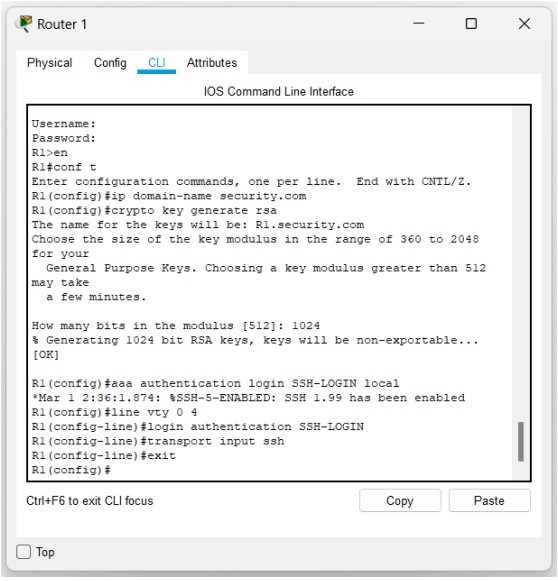
Username is admin123 and password is ty23.



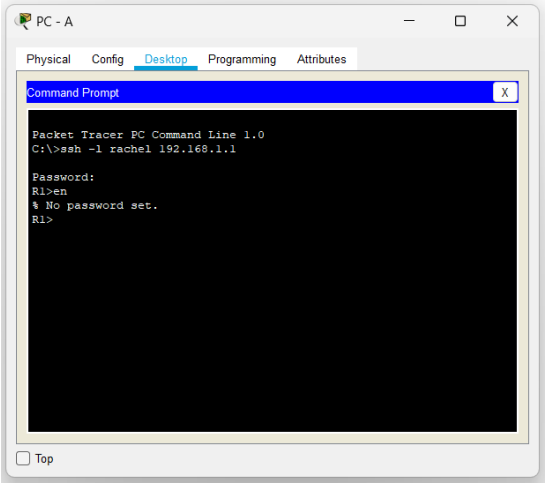
E) Configure local AAA authentication for vty lines on Router 1

From security.com it is

Router 1
r1>en
r1#conf t
r1(config)#ip domain-name security.com
r1(config)#crypto key generate rsa
How many bits in the modulus will be asked, enter 1024.
r1(config)#aaa authentication login SSH-LOGIN local
r1(config)#line vty 0 4
r1(config-line)#login authentication SSH-LOGIN
r1(config-line)#transport input ssh
r1(config-line)#exit

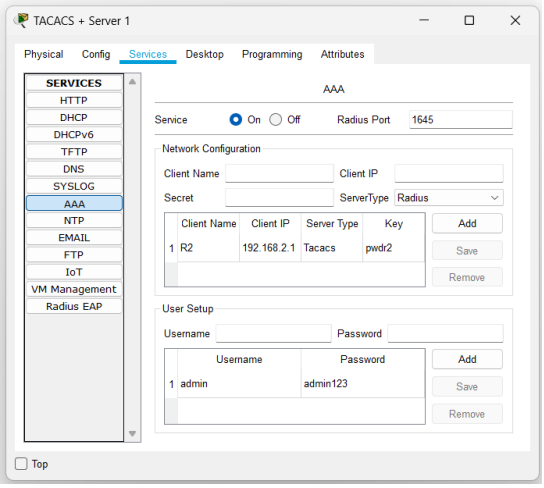


Go to PC – A Command Prompt and enter ssh -l admin123 192.168.1.1. It will ask for password, enterty23.



F) Configure service-based AAA authentication using TACACS+ on Router 2

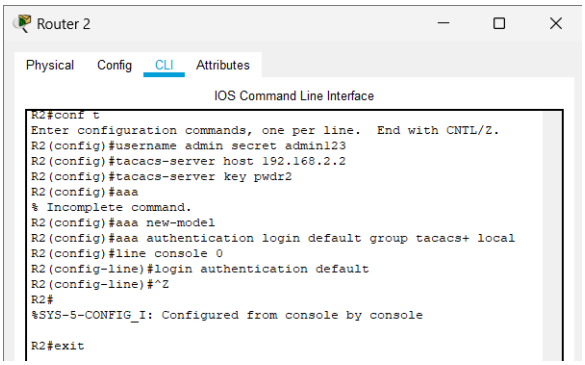
Go to services of TACACS+ Server and then select AAA. Switch on the service and enter the client IP and password. Setup username and password as well.



<u>Router 2</u>

```
r2>en
r2#conf t
r2(config)#username admin secret admin123
r2(config)#tacacs-server host 192.168.2.2
r2(config)#tacacs-server key pwdr2
r2(config)#aaa new-model
r2(config)#aaa authentication login default group tacacs+ local
r2(config)#line console 0
r2(config-line)#login authentication default
```

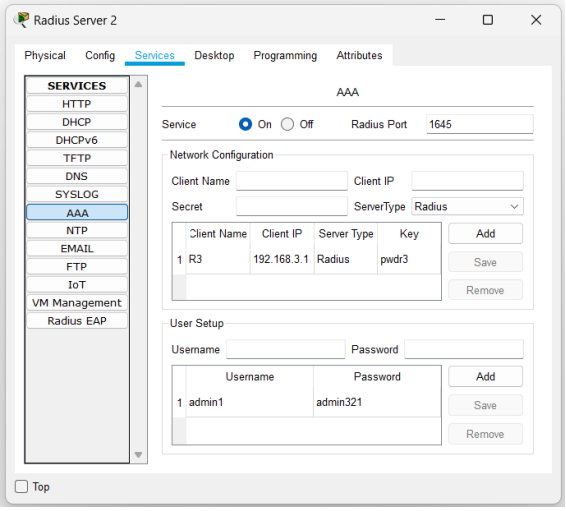
User Access Verification is asked enter the credentials of the TACACS+ Server.



Username is admin and password is admin123 which is also the username and password set for TACACS Server.

G) Configure server-based AAA authentication using RADIUS on Router 3

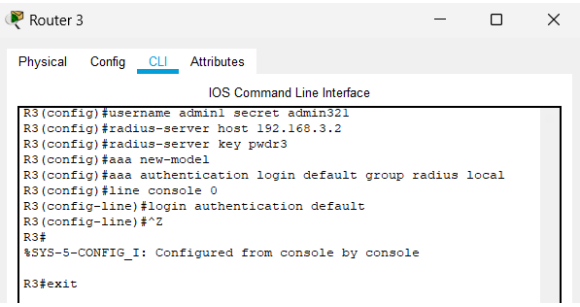
Go to services of RADIUS Server and then select AAA. Switch on the service and enter the client IP and password. Setup username and password as well.

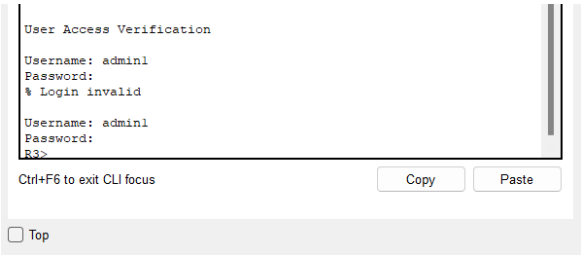


<u>Router 3</u>

```
r3>en
r3#conf t
r3(config)#username admin1 secret admin321
r3(config)#radius-server host 192.168.3.2
r3(config)#radius-server key pwdr3
r3(config)#aaa new-model
r3(config)#aaa authentication login default group radius local
r3(config)#line console 0
r3(config-line)#login authentication default
```

User Access Verification is asked enter the credentials of the RADIUS Server.





Username is admin1 and password is admin321 which is also set in radius server.

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Practical 11: Configure a Zone-Based Policy Firewall (ZPF).

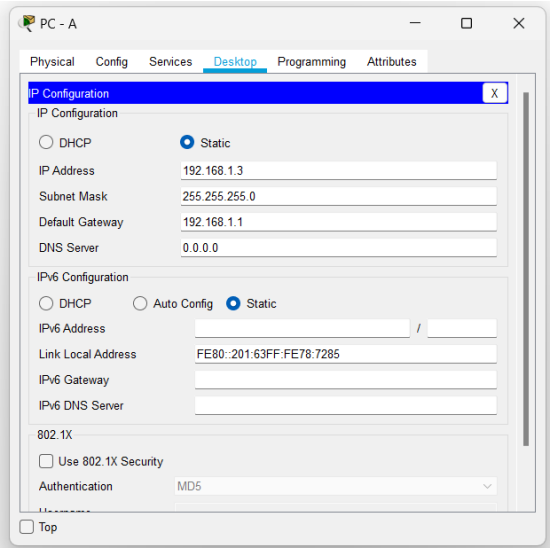
A) Topology

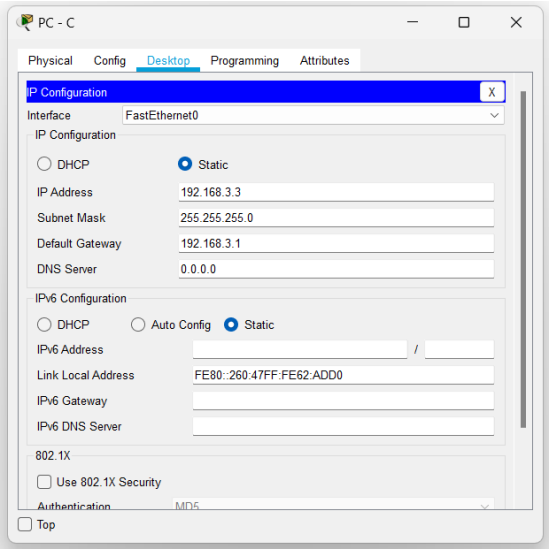
1. Drag and drop the required icons that are 3 1941 Routers; 1 PT-PC; 2 2960-24TT Switches; 1 PT-Server and rename them accordingly.

2. Define the respective IP configuration by going to Desktop and then to IP Configuration for PC - A(Server) and PC - C.

Addressing Table

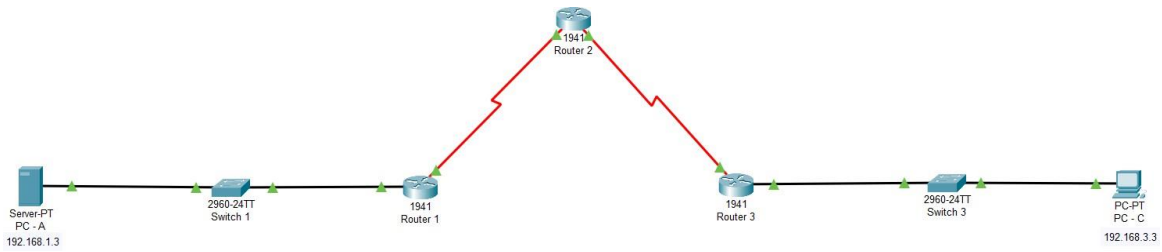
Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18





3. Add serial ports to all the three routers by going to Physical of Router 1, switching the router off and adding HWIC-2T to both the empty slots and turn the switch on, do the same for Router 2 and Router 3.

4. Connect the devices with the respective wired connection.



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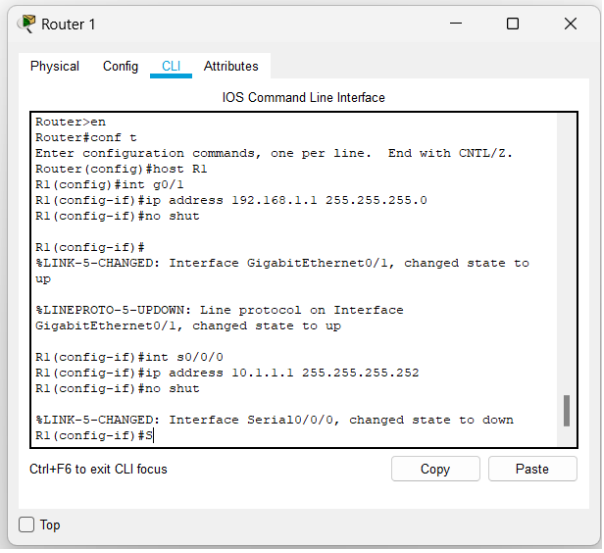
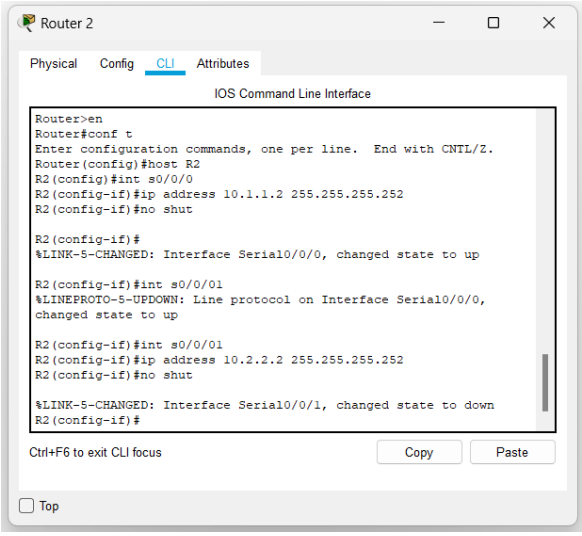
B) Configure Router

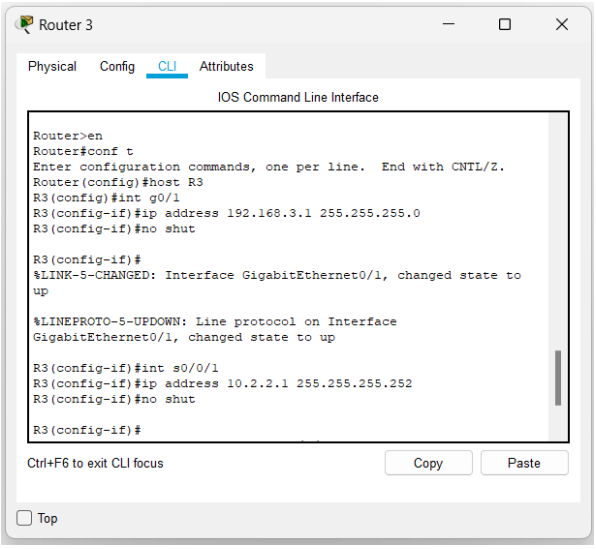
<u>Router 1</u>
Router>en
Router#conf t
Router(config)#host r1
r1(config)#int g0/1
r1(config-if)# ip address 192.168.1.1 255.255.255.0
r1(config-if)#no shut
r1(config-if)#int s0/0/0
r1(config-if)#ip address 10.1.1.1 255.255.255.252
r1(config-if)#no shut



<u>Router 2</u>
Router>en
Router#conf t
Router(config)#host r2
r2(config)#int s0/0/0
r2(config-if)#ip address 10.1.1.2 255.255.255.252
r2(config-if)#no shut
r2(config-if)#int s0/0/1
r2(config-if)#ip address 10.2.2.2 255.255.255.252
r2(config-if)#no shut



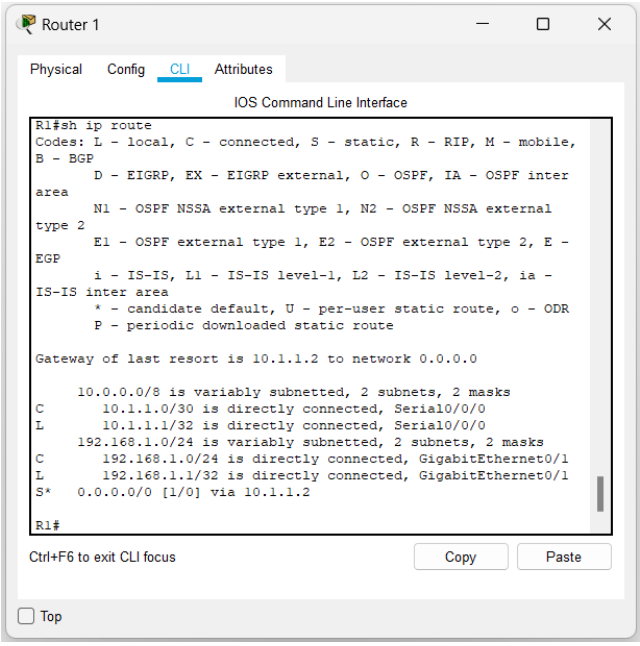
<u>Router 3</u>
Router>en
Router#conf t
Router(config)#host r3
r3(config)#int g0/1
r3(config-if)#ip address 192.168.3.1 255.255.255.0
r3(config-if)#no shut
r3(config-if)#int s0/0/1
r3(config-if)#ip address 10.2.2.1 255.255.255.252
r3(config-if)#no shut



C) Configure Static Routing

Go to Config of Router 1 → Static under Routing and add the details that are
Network: 0.0.0.0
Mask: 0.0.0.0
Next Hop: 10.1.1.2

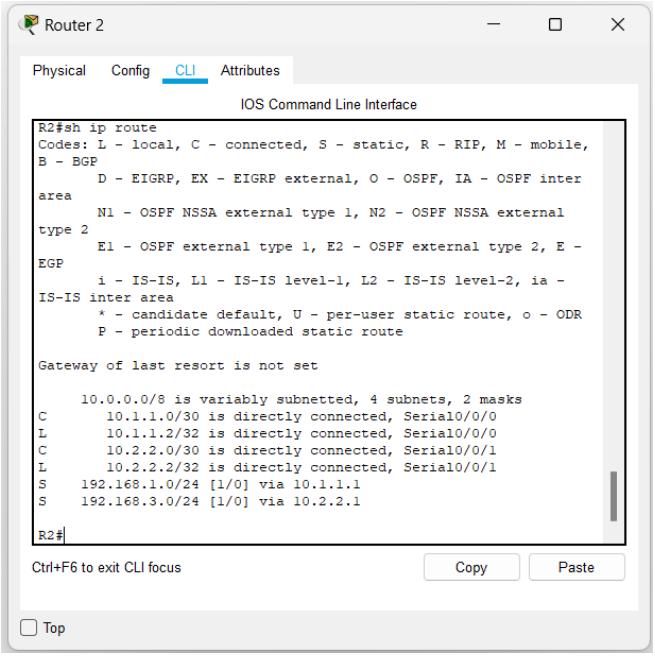
r1#show ip route



Router 2
Network: 192.168. 1.0
Mask: 255.255.255.0
Next Hop: 10.1.1.1

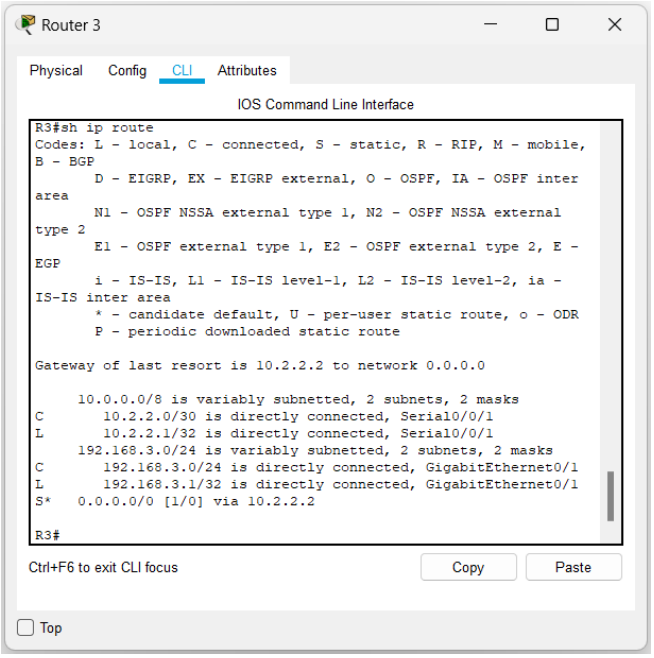
Network: 192.168.3.0
Mask: 255.255.255.0
Next Hop: 10.2.2.1

r2#show ip route



Router 3
Network: 0.0.0.0
Mask: 0.0.0.0
Next Hop: 10.2.2.2

r3#show ip route

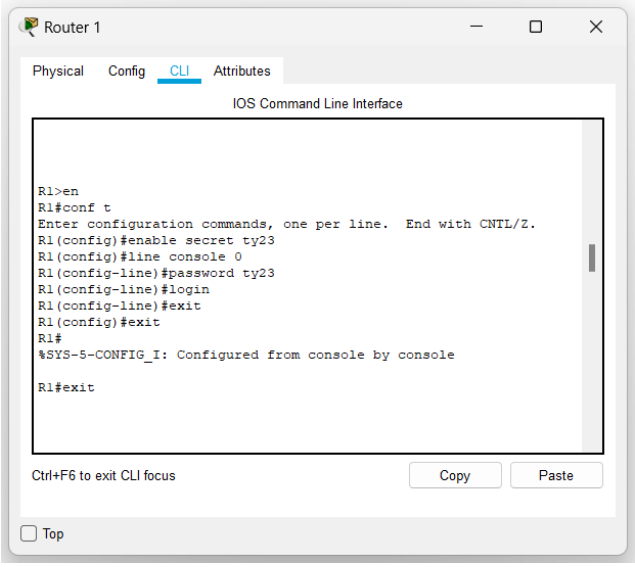


The S proves that static routing is successful.

D) Password Authentication of Router

Router 1

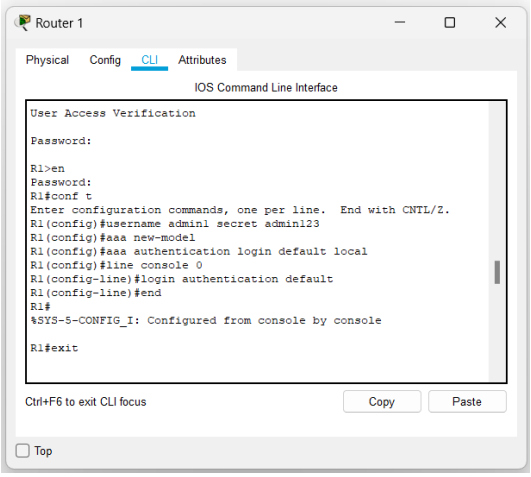
r1(config)#enable secret ty23
r1(config)#line console 0
r1(config-line)#password ty23
r1(config-line)#login
r1(config-line)#exit



Exit from the connection.
Under User Access Verification password is asked, enter ty23, it will not be seen but the system will accept it.
Connection to the router will open again.

Router 1

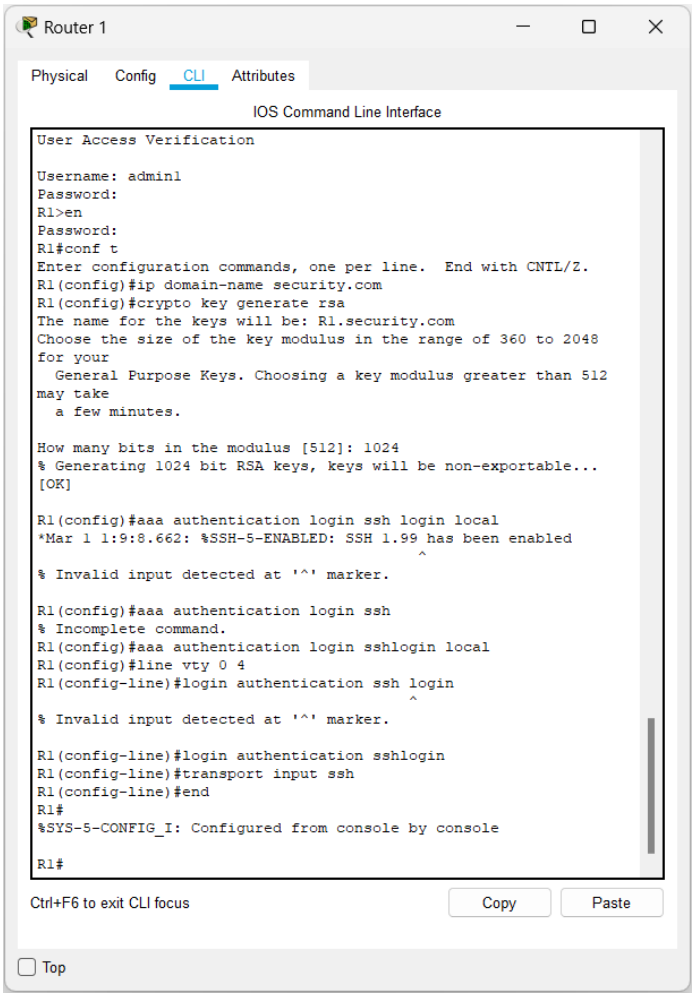
r1#conf t
r1(config)#username admin1 secret admin123
r1(config)#aaa new-model
r1(config)#aaa authentication login default local
r1(config)#line console 0
r1(config-line)#login authentication default
r1(config-line)#end



End the connection completely.
Again, User Access Verification will come, enter Username as admin1 and Password as admin123.
After enabling the router if again a password is asked enter ty23.

Router 1

r1(config)#ip domain-name security.com
r1(config)#crypto key generate rsa
How many bits in the modulus will be asked, enter 1024.
r1(config)#aaa authentication login sshlogin local
r1(config)#line vty 0 4
r1(config-line)#login authentication sshlogin
r1(config-line)#transport input ssh
r1(config-line)#end

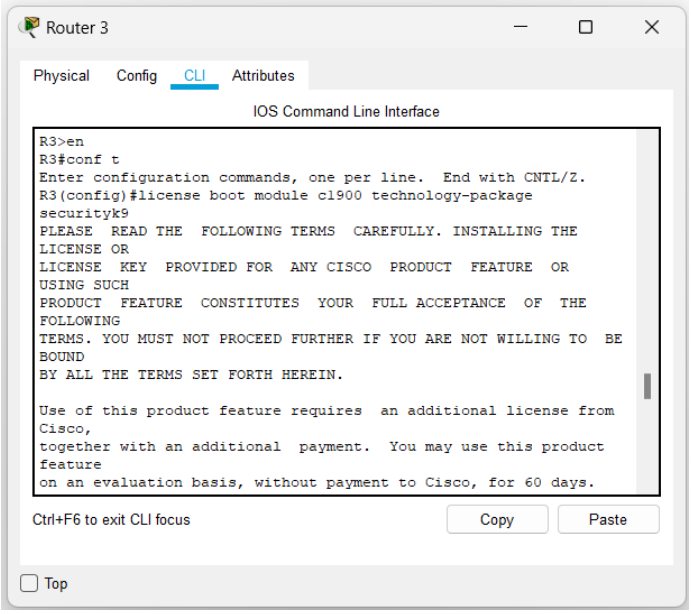


E) Enable security technology package on Router3

Router 3

r3(config)#license boot module c1900 technology-package securityk9

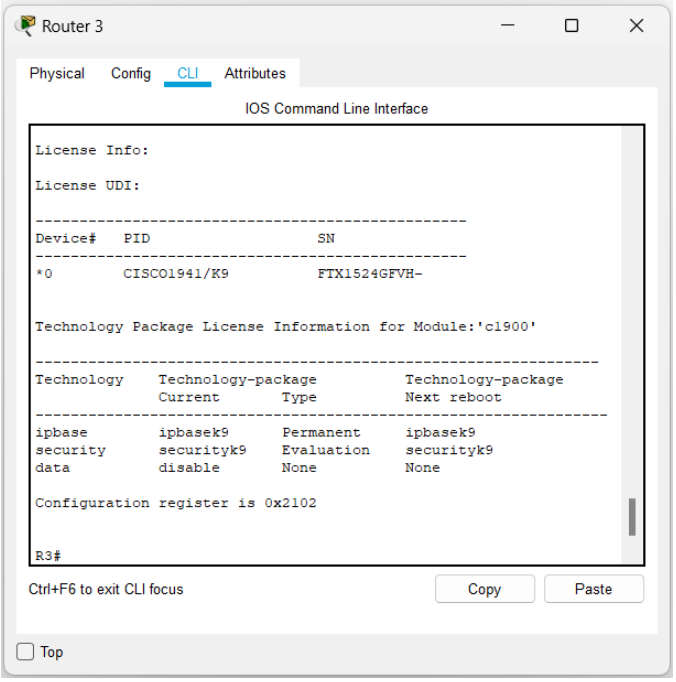
Accept the agreement by typing yes.



r3#reload

Confirm the reload.

r3#show version



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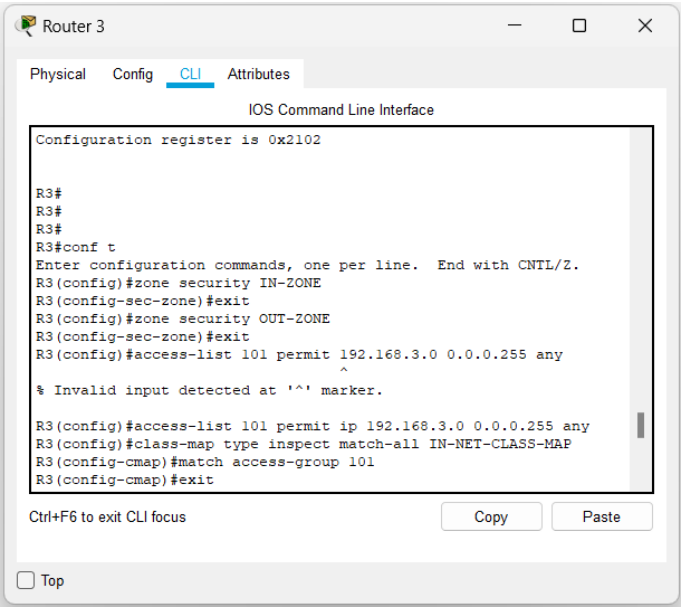
F) Create firewall zone on Router3

r3(config)#zone security IN-ZONE
r3(config-sec-zone)#exit
r3(config)#zone security OUT-ZONE
r3(config-sec-zone)#exit

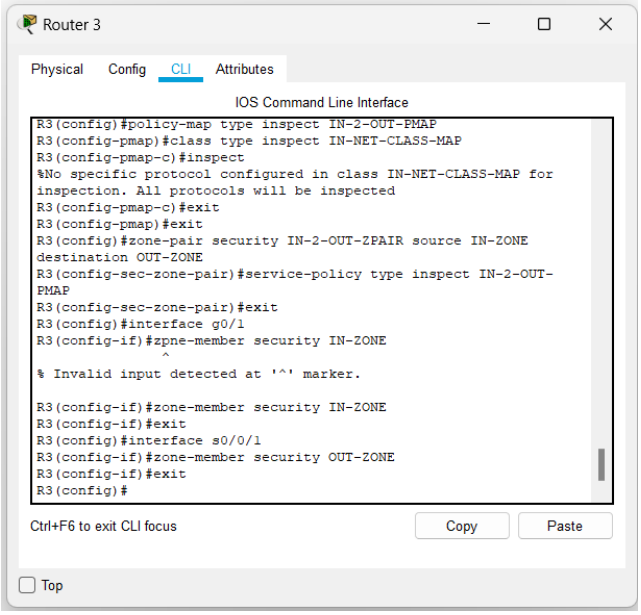
G) Identify traffic using a class-map

r3(config)# access-list 101 permit ip
192.168.3.0 0.0.0.255 any
r3(config)#
class-map type inspect match-all IN-NET-CLASS-MAP
r3(config-cmap)# match access-group 101
r3(config-cmap)# exit



H) Firewall policies

r3(config)# policy-map type inspect IN-2-OUT-PMAP
r3(config-pmap)# class type inspect IN-NET-CLASS-MAP
r3(config-pmap-c)# inspect
r3(config-pmap-c)# exit
r3(config-pmap)# exit
r3(config)# zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUTZONE
r3(config-sec-zone-pair)# service-policy type inspect IN-2-OUT-PMAP
r3(config-sec-zone-pair)# exit
r3(config)# interface g0/1
r3(config-if)# zone-member security IN-ZONE
r3(config-if)# exit
r3(config)# interface s0/0/1
r3(config-if)# zone-member security OUT-ZONE
r3(config-if)# exit



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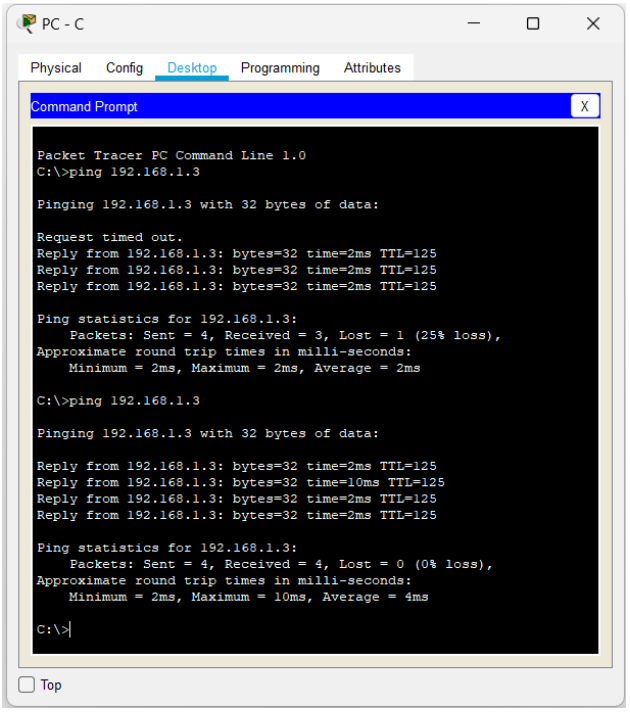
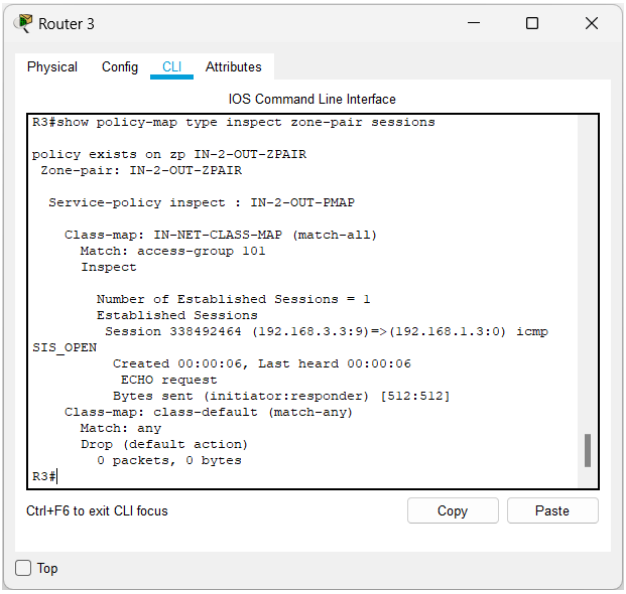
I) Test firewall functionality from IN-ZONE to OUT-ZONE

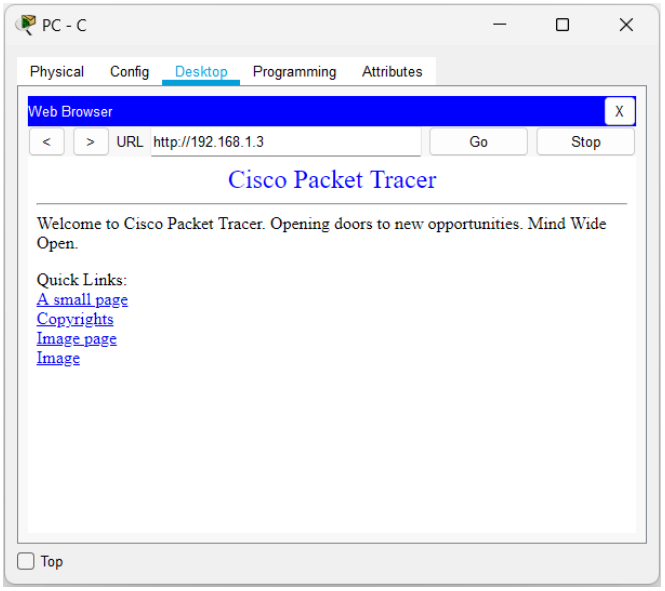
Open Command Prompt of PC – C and ping 192.168.1.3 that is PC – A (Server). It should be successful.

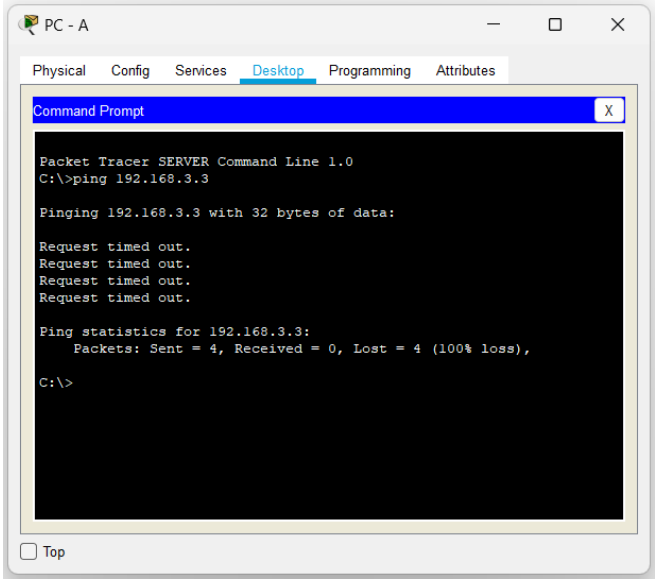


In CLI mode of Router 3 type,
r3#show policy-map type inspect zone-pair sessions
The details regarding the policy should be displayed.



Go to Web Browser of PC - C enter http://192.168.1.3 and click on Go, Cisco Packet Tracer page will be displayed.



J) Test firewall functionality from OUT-ZONE to IN-ZONE

Open Command Prompt of PC – A and ping 192.168.3.3 that is PC – C. It should fail.